

Math 871 Topology I

Problem Set 6

1. Let X and Y be topological spaces and $f: X \rightarrow Y$ be an injective continuous function.
 - (a) Prove that if Y is Hausdorff then X is Hausdorff.
 - (b) Does the converse of (a) hold?

2. Let X be a Hausdorff space.

- (a) Show that if $f: X \rightarrow X$ is continuous, then

$$C := \{x \in X \mid f(x) = x\}$$

is closed.

- (b) Let $A \subseteq X$ have the subspace topology. Show that if there exists a retraction $r: X \rightarrow A$ of X onto A , then A is a closed subset of X .
 - (c) Does (b) hold if we do not assume that X is Hausdorff?